

New Attempt

- Due Dec 9, 2024 by 2:59am
- Points 50
- Submitting a text entry box, a website url, or a file upload

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## Robot Project

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### Robot Project Individual Portfolio

#### Overview:

A portfolio is a chance for you to (1) practice your engineering documentation skills, (2) begin compiling a record of your engineering projects for co-op and job applications, and (3) to reflect on your engineering design process.

#### Tasks:

**Using the design history documented in your team's digital design notebook and the project descriptions provided to you on Canvas, create a portfolio in the form of a personal website or PDF that contains the followings:**

- **Introduction**
  - Write an introduction that provides an overview of your project.
  - Include the purpose, context, and any constraints or goals associated with the project.
- **Define the Problem**
  - Describe the initial problem or challenge you aimed to solve.
  - Reference early sketches, brainstorming sessions, or design requirements documented in your digital design notebook.
- **Document Your Process**
  - Highlight key milestones in your design process using entries from your design notebook.
  - Include screenshots or photos of sketches, assembly, flowcharts, and wire diagrams to illustrate the progression.
  - Include videos of your robot completing each challenge
  - Include photos and/or videos of your modular maze.
  - AutoCAD drawing(s) of your modular maze system and/or components of your robot (at least one AutoCAD drawing)

- If you made custom parts (laser cut and/or 3D printed) for your robot, you should definitely include drawings and photos of those!
- **Explore Iterations**
  - A story about the iterations you made and the process you followed along the way.
  - Your thought process as you worked on the different challenges
  - What you tried and didn't work
  - What you tried and worked
  - What was unique or different about your robot
  - Share how feedback or challenges influenced changes to your design, supported by evidence from your design history.
- **Share Challenges and Solutions**
  - Identify major challenges encountered during the project from technical, construction, and/or team work perspectives
  - Describe how you addressed these challenges, using specific examples from your notebook entries.
- **Showcase the Final Design**
  - Present your final design with detailed visuals (e.g., renders, photos, technical drawings).
  - Explain how the final product addresses the initial problem and meets the project goals.
- **Add Personal Reflections**
  - Reflect on what you learned during the project and how your skills have developed.
  - Share insights into what you might do differently in the future.
  - Talk about your specific contribution to the group project.
- **Create a Narrative**
  - Organize your portfolio to tell a clear and compelling story about your project.
  - Ensure there is a logical flow from identifying the problem to showcasing the final design.
- **Include Citations**
  - Credit any external sources, tools, or collaborators that contributed to your project.
- **Polish Your Presentation**
  - Review your portfolio for clarity, professionalism, and visual appeal.
  - Ensure that all images, captions, and text are cohesive and align with your narrative.
- **Ethical Considerations in a Robot Project**

- Reflecting on the lecture on *Technology and Society*, analyze the societal impact of your autonomous robot or another autonomous robot of your choice. Discuss the new forms of life it could create, highlighting both the positive and negative outcomes. Additionally, explore how it might influence power dynamics, focusing on aspects of social justice or potential inequities it may introduce or exacerbate.
- Don't forget to cite any sources you used for this section

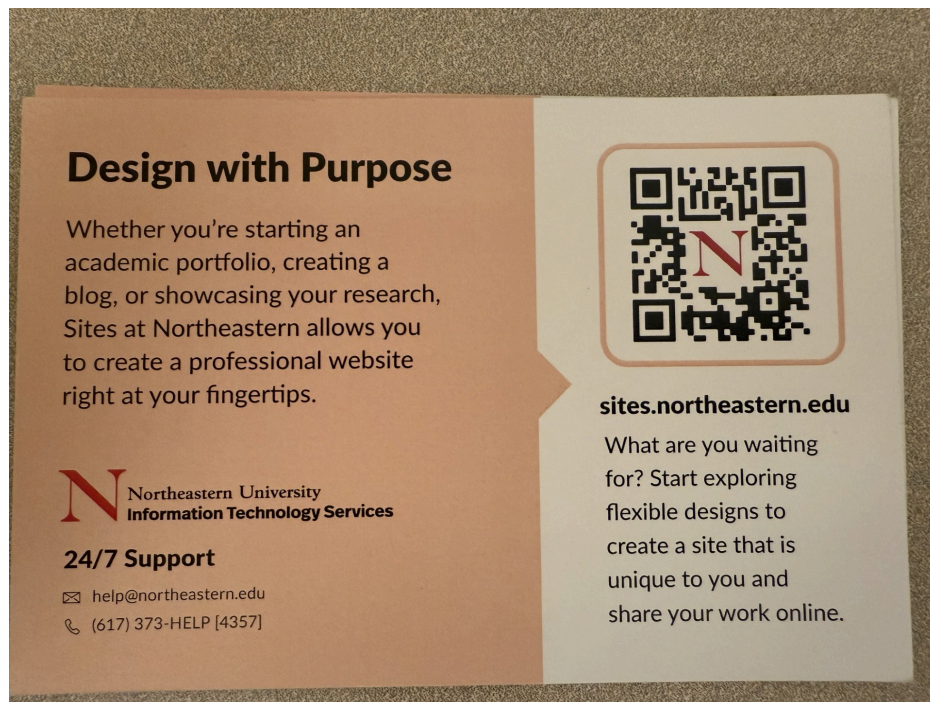
**Outside of these requirements, you have A LOT of flexibility in how you construct your portfolio! Most importantly, the portfolio is supposed to represent YOU. It can be in your style and tell the story of this project in whatever way makes sense to you.**

**Make sure you grant access to all the links so that other people will be able to view them! Check that everything is working using incognito mode on the chrome browser. Refer to the rubric below for a full breakdown of how the portfolio will be evaluated.**

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## Resources

1. You may start building your website by going to [sites.northeastern.edu](https://sites.northeastern.edu). NU ITS provides 24/7 help on designing your own website. Refer to the following flyer.



2. Here's some advice from the MIT Comm Lab that you may find helpful: <https://mitcommlab.mit.edu/meche/commkit/portfolio/>  
(<https://mitcommlab.mit.edu/meche/commkit/portfolio/>)

- A nice example from MIT: <https://www.hilaryannajohnson.com/>  
(<https://www.hilaryannajohnson.com/>)
3. Example of a PDF style portfolio:
- <https://pachecoandre.com.br/assets/files/andre-pacheco-eng-undergrad-portfolio.pdf>  
(<https://pachecoandre.com.br/assets/files/andre-pacheco-eng-undergrad-portfolio.pdf>)
4. Examples of websites from Northeastern students (from former students and/or simple google search):
- <https://aspenhand.sites.northeastern.edu/> ↗ (<https://aspenhand.sites.northeastern.edu/>)
  - <https://sites.google.com/view/lily-p-engineering/robot-project> ↗  
(<https://sites.google.com/view/lily-p-engineering/robot-project>)
  - <https://johnduncanstevens.wixsite.com/portfolio>  
(<https://johnduncanstevens.wixsite.com/portfolio>)
  - <https://www.max-berman.com/> (<https://www.max-berman.com/>)
  - <https://jasneengineeringportfolio.weebly.com/> (<https://jasneengineeringportfolio.weebly.com/>)
  - <https://www.vishakht.com/projects> ↗ (<https://www.vishakht.com/projects>)
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## File submission

- **Submit your PDF file or a link to your website**
- **Final To-Do List for Cornerstone 1:** Here's everything you need to complete:
  - **Complete any late assignments** (due 12/4 by midnight).
  - **After Challenge 4 Demo:** Take apart your robot. Disassemble the electronics from your robot chassis and return the nuts and bolts. No need to take apart the chassis itself. You may keep your robot chassis or give it back to us for recycling/reuse. (due 12/5).
  - **Clean out your team's cabinet shelf** (due 12/5).
  - **Organize your SparkFun kit:**
    - Reassemble your kit and keep it safe for next semester.
    - If you're heading to London, make sure to take it with you!
    - Return any extra components to the classroom so they can be redistributed.
    - Let me know if you think your kit got switched with someone else's.
  - **Return borrowed items** (due 12/5):

- This includes measuring tapes, rulers, calipers, USB-C converters, or anything else borrowed ("or borrowed").
- **Submit your portfolio** (due 12/8 by midnight).
- **Complete TRACE evaluations:** Please share your feedback on what worked and what didn't this semester! Cornerstone has evolved over the past 10 years thanks to input from students like you. Your thoughts can help future students succeed.

## Portfolio Rubric

| Criteria                                                                                                                                                                                                                                                                                                                          | Ratings                                                                      | Pts           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------|
| <p>Technical Proficiency</p> <p>This category reflects how your portfolio demonstrates your technical skills and knowledge. I will evaluate the accuracy, completeness, and quality of the technical content within your portfolio, including flowcharts, AutoCAD drawings, and any other technical documentation.</p>            | <p><b>10 pts<br/>Full<br/>Marks</b></p> <p><b>0 pts<br/>No<br/>Marks</b></p> | <p>10 pts</p> |
| <p>Presentation/Communication/Style</p> <p>This category focuses on the overall presentation, organization, clarity, and style of the your portfolio. I will evaluate how well you have communicated your work, whether your portfolio is engaging and professional, and if the content is easy to follow.</p>                    | <p><b>10 pts<br/>Full<br/>Marks</b></p> <p><b>0 pts<br/>No<br/>Marks</b></p> | <p>10 pts</p> |
| <p>Following Requirements</p> <p>I will evaluate whether you have included all required elements in your portfolio: flowcharts, photos, videos, AutoCAD drawings, the story of your design process etc.</p>                                                                                                                       | <p><b>10 pts<br/>Full<br/>Marks</b></p> <p><b>0 pts<br/>No<br/>Marks</b></p> | <p>10 pts</p> |
| <p>Engineering Design Process</p> <p>This category focuses on the story of your engineering design process. I will be looking for evidence of the design process phases that we discussed in class and in the textbook readings (defining the problem, generating solutions, decision-making, evaluation, iteration).</p>         | <p><b>10 pts<br/>Full<br/>Marks</b></p> <p><b>0 pts<br/>No<br/>Marks</b></p> | <p>10 pts</p> |
| <p>Reflection</p> <p>This category focuses specifically on the depth of reflection within your portfolio. I will be looking for analysis of your design process, discussion of what worked and what didn't, discussion of lessons learned, and reflection on your overall project experience. and contribution to the project</p> | <p><b>10 pts<br/>Full<br/>Marks</b></p> <p><b>0 pts<br/>No<br/>Marks</b></p> | <p>10 pts</p> |
| <p>Total Points: 50</p>                                                                                                                                                                                                                                                                                                           |                                                                              |               |